	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^-}^{\#1\,\alpha\beta\chi}$
$\mathcal{K}_{0^+}^{\#1\,\dagger}$	0	0
$\mathcal{K}_{0^-}^{\#1\,\dagger\alpha\beta\chi}$	0	0

	$\mathcal{K}_{1^+}^{\#1\,\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2\,\alpha\beta}$	$\mathcal{K}_{1^-}^{\#1\,\alpha}$	$\mathcal{K}_{1^-}^{\#2\,\alpha}$
$\mathcal{K}_{1^+}^{\#1\,\dagger\alpha\beta}$	$-\frac{\Upsilon_1 k^2}{6}$	$\frac{\Upsilon_1 k^2}{3\sqrt{2}}$	0	0
$\mathcal{K}_{1^+}^{\#2\,\dagger\alpha\beta}$	$\frac{\Upsilon_1 k^2}{3\sqrt{2}}$	$-\frac{\Upsilon_1 k^2}{3}$	0	0
$\mathcal{K}_{1^-}^{\#1\,\dagger\alpha}$	0	0	$-k^2\,{}^{(4)}\kappa_2$	$\frac{k^2\,{}^{(4)}\kappa_2}{\sqrt{2}}$
$\mathcal{K}_{1^-}^{\#2\,\dagger\alpha}$	0	0	$\frac{k^2\,{}^{(4)}\kappa_2}{\sqrt{2}}$	$-\frac{k^2\,{}^{(4)}\kappa_2}{2}$

	$\mathcal{K}_{2^+}^{\#1\,\alpha\beta}$	$\mathcal{K}_{2^-}^{\#1\,\alpha\beta\chi}$
$\mathcal{K}_{2^+}^{\#1\,\dagger\alpha\beta}$	$\frac{3\Upsilon_1 k^2}{2}$	0
$\mathcal{K}_{2^-}^{\#1\,\dagger\alpha\beta\chi}$	0	0

	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^-}^{\#1\,\alpha\beta\chi}$
$\mathcal{J}_{0^+}^{\#1\,\dagger}$	0	0
$\mathcal{J}_{0^-}^{\#1\,\dagger\alpha\beta\chi}$	0	0

	$\mathcal{J}_{1^+}^{\#1\,\alpha\beta}$	$\mathcal{J}_{1^+}^{\#2\,\alpha\beta}$	$\mathcal{J}_{1^-}^{\#1\,\alpha}$	$\mathcal{J}_{1^-}^{\#2\,\alpha}$
$\mathcal{J}_{1^+}^{\#1\,\dagger\alpha\beta}$	$\frac{1}{3\,\mathrm{Det}(1^+)}$	$-\frac{\sqrt{2}}{3\,\mathrm{Det}(1^+)}$	0	0
$\mathcal{J}_{1^+}^{\#2\,\dagger\alpha\beta}$	$-\frac{\sqrt{2}}{3\,\mathrm{Det}(1^+)}$	$\frac{2}{3\,\mathrm{Det}(1^+)}$	0	0
$\mathcal{J}_{1^-}^{\#1\,\dagger\alpha}$	0	0	$-\frac{4}{9k^2\,{}^{(4)}\kappa_2}$	$\frac{2\sqrt{2}}{9k^2\,{}^{(4)}\kappa_2}$
$\mathcal{J}_{1^-}^{\#2\,\dagger\alpha}$	0	0	$\frac{2\sqrt{2}}{9k^2\,{}^{(4)}\kappa_2}$	$-\frac{2}{9k^2\,{}^{(4)}\kappa_2}$

	$\mathcal{J}_{2^+}^{\#1\,\alpha\beta}$	$\mathcal{J}_{2^-}^{\#1\,\alpha\beta\chi}$
$\mathcal{J}_{2^+}^{\#1\,\dagger\alpha\beta}$	$\frac{1}{\mathrm{Det}(2^+)}$	0
$\mathcal{J}_{2^-}^{\#1\,\dagger\alpha\beta\chi}$	0	0

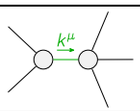
Abbreviations used in matrices

$$\Upsilon_1 == {}^{(4)}\kappa_1 + {}^{(4)}\kappa_2 \;\&\&\; \mathrm{Det}(1^+) == -\frac{1}{2}k^2({}^{(4)}\kappa_1 + {}^{(4)}\kappa_2) \;\&\&\; \mathrm{Det}(2^+) == \frac{3}{2}k^2({}^{(4)}\kappa_1 + {}^{(4)}\kappa_2)$$

Lagrangian

$$\begin{aligned}&{}^{(4)}\kappa_1\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + {}^{(4)}\kappa_2\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} - \frac{4}{3}\,{}^{(4)}\kappa_1\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} - \\ &\frac{4}{3}\,{}^{(4)}\kappa_2\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} + \frac{2}{3}\,{}^{(4)}\kappa_1\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} + \frac{2}{3}\,{}^{(4)}\kappa_2\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} - \\ &\frac{5}{3}\,{}^{(4)}\kappa_1\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - \frac{5}{3}\,{}^{(4)}\kappa_2\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - \frac{1}{3}\,{}^{(4)}\kappa_1\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} - \frac{1}{3}\,{}^{(4)}\kappa_2\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi}\end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\;\mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}_{0^-}^{\#1\,\alpha\beta\chi} == 0$	1	$\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + \partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + \partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + \partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} == \partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + \partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + \partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}$
$\mathcal{J}_{0^+}^{\#1} == 0$	1	$\partial_\beta\mathcal{J}^{\alpha}{}_{\alpha}{}^{\beta} == 0$
$\mathcal{J}_{1^-}^{\#1\,\alpha} + 2\,\mathcal{J}_{1^-}^{\#2\,\alpha} == 0$	3	$\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta} == 3\,\partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi}$
$2\,\mathcal{J}_{1^+}^{\#1\,\alpha\beta} + \mathcal{J}_{1^+}^{\#2\,\alpha\beta} == 0$	3	$\partial_\chi\mathcal{J}^{\alpha\beta\chi} + \partial_\chi\mathcal{J}^{\chi\alpha\beta} == \partial_\chi\mathcal{J}^{\beta\alpha\chi}$
$\mathcal{J}_{2^-}^{\#1\,\alpha\beta\chi} == 0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}{}_{\delta} == 3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}{}_{\delta}$
Total # constraint(s):	13	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{1}{{}^{(4)}\kappa_2}$
Resolved unitarity condition(s):	${}^{(4)}\kappa_2 > 0$		